BEST OBSERVED SETTING FOR EACH METRIC

No single method–budget combination wins every metric

CCC

Similarity

K = 10

0.096

+7.5% vs baseline

Agreement in trend, mean and variance

PCC(M)

ST Hard

K = 5

0.217

+3.2% vs baseline

Mean correlation across 250 genes

PCC(H)

Similarity

K = 5

0.381

+1.9% vs baseline

Mean correlation across top-50 genes

MSE

Similarity

K = 15

1.344

+1.6% vs baseline

Large errors count more

MAE

Similarity

K = 15

0.890

+0.9% vs baseline

Average absolute prediction error

Interpretation: K=5 gives the strongest PCC(M) and PCC(H); Similarity K=15 gives the lowest errors.